Netaji Subhash Engineering College

PRESENTATION ON :- **Classification using ANN :-**

DATA WAREHOUSING AND DATA MINING PEC-IT602B(6584)

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Data Warehousing and Data Mining:

Classification Using Artificial Neural Networks (ANN)

Unlocking insights from complex data with intelligent algorithms.

What is Data Warehousing and Data Mining?

Data Warehousing

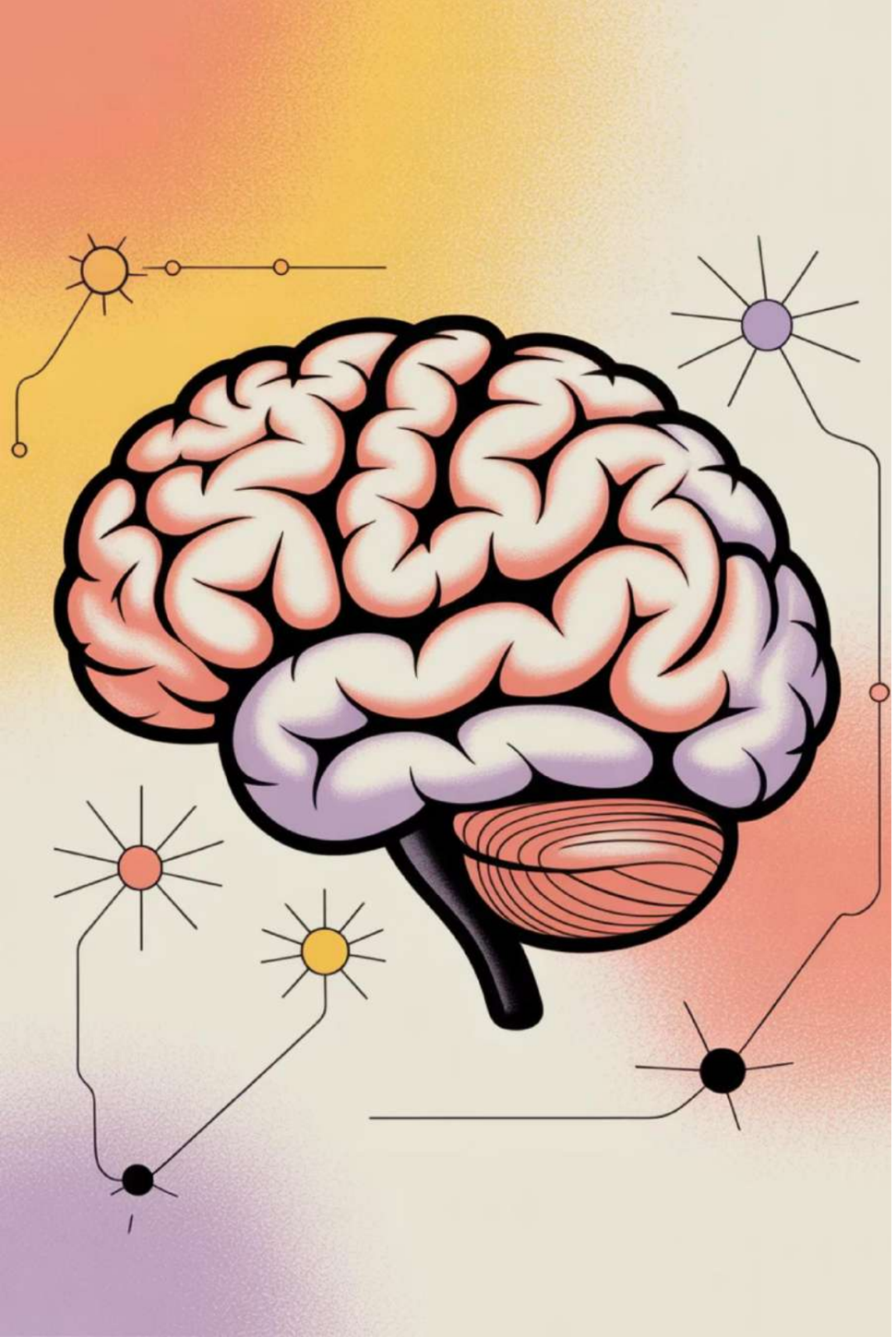
A robust centralized repository for vast volumes of historical data, consolidated from various operational sources, designed for query and analysis rather than transaction processing.

Data Mining

The systematic process of extracting meaningful patterns, trends, and knowledge from large datasets, transforming raw information into actionable intelligence.

Classification

A fundamental data mining technique that categorizes data points into predefined classes or labels, crucial for predictive modeling and decision-making.



Introduction to Artificial Neural Networks (ANN)

Artificial Neural Networks (ANNs) are computational models inspired by the biological neural networks that constitute animal brains.

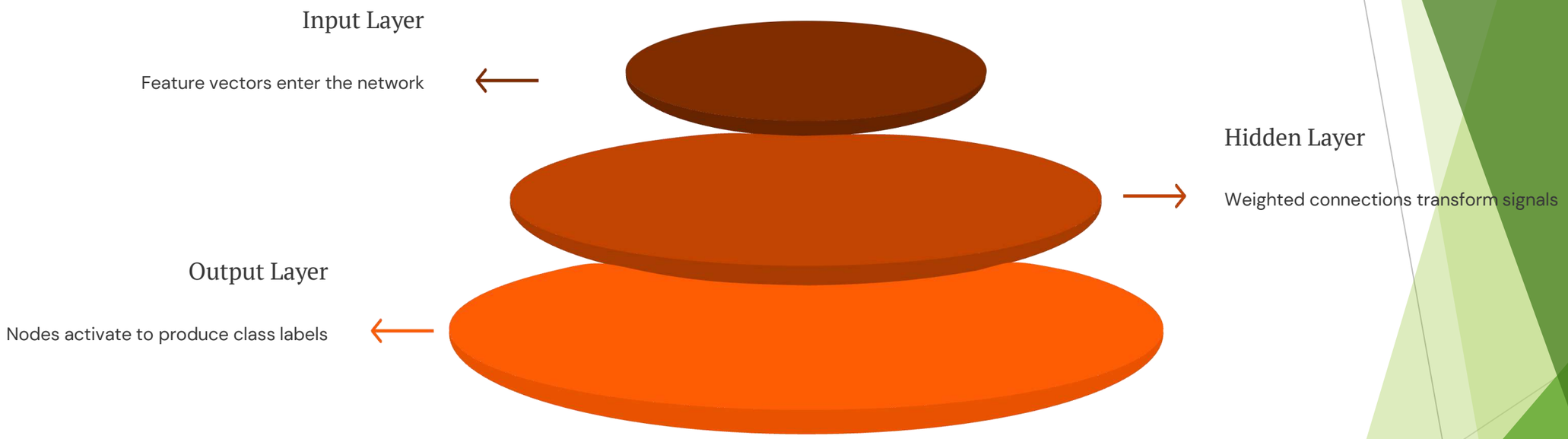
1. **Brain-Inspired**
   Mimics the intricate structure and function of the human brain's neurons.

2. **Interconnected Processors**
   Comprises artificial neurons linked to each other, processing and transmitting information.

3. **Nonlinear Learning**
   Excels at discovering complex, non-linear relationships within vast datasets.

4. **Robust Classification**
   Highly effective for classification tasks in data mining due to its adaptability and precision.

How ANN Performs Classification

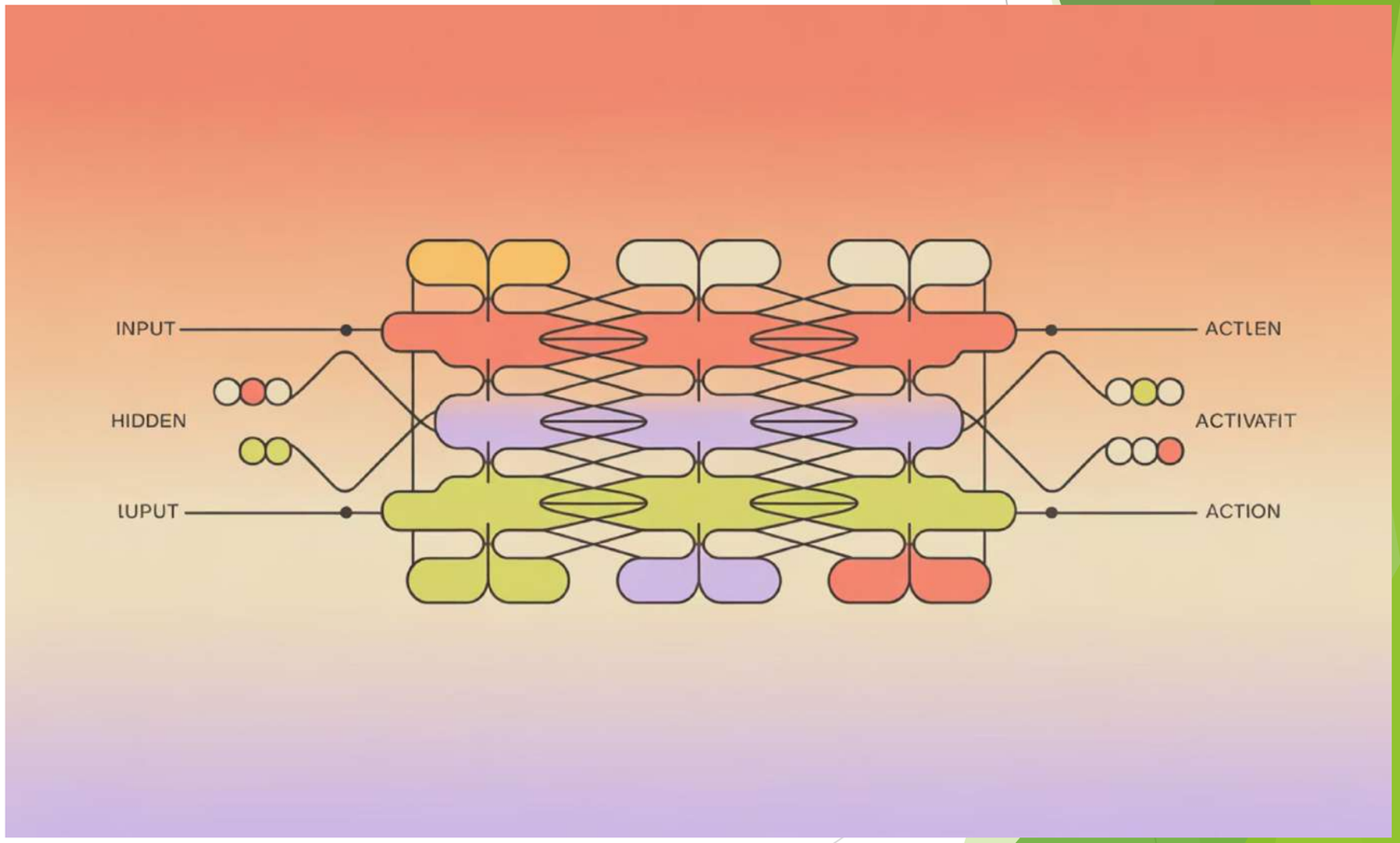


ANNs classify by processing input features through weighted connections across multiple layers, ultimately activating output nodes corresponding to specific class labels.

ANN Architecture & Working

The most prevalent ANN architecture in data mining is the feedforward multilayer network, where data flows in one direction from input to output. Each neuron aggregates weighted inputs and a bias, then applies an activation function (e.g., sigmoid, ReLU) to produce an output.

Training involves algorithms like backpropagation, which adjust weights and biases iteratively to minimize the discrepancy between predicted and actual outputs. Functional Link ANN (FLANN) variants simplify the architecture, often leading to faster convergence and improved performance for specific classification challenges.



Why Use ANN for Classification in Data Mining?

ANNs are powerful tools for data mining classification, offering unique advantages in handling real-world data challenges.

1

Handles Noisy Data

Effectively processes datasets with imperfections, missing values, or outliers.

2

Complex Relationships

Capable of discerning intricate, non-linear patterns that traditional methods might miss.

3

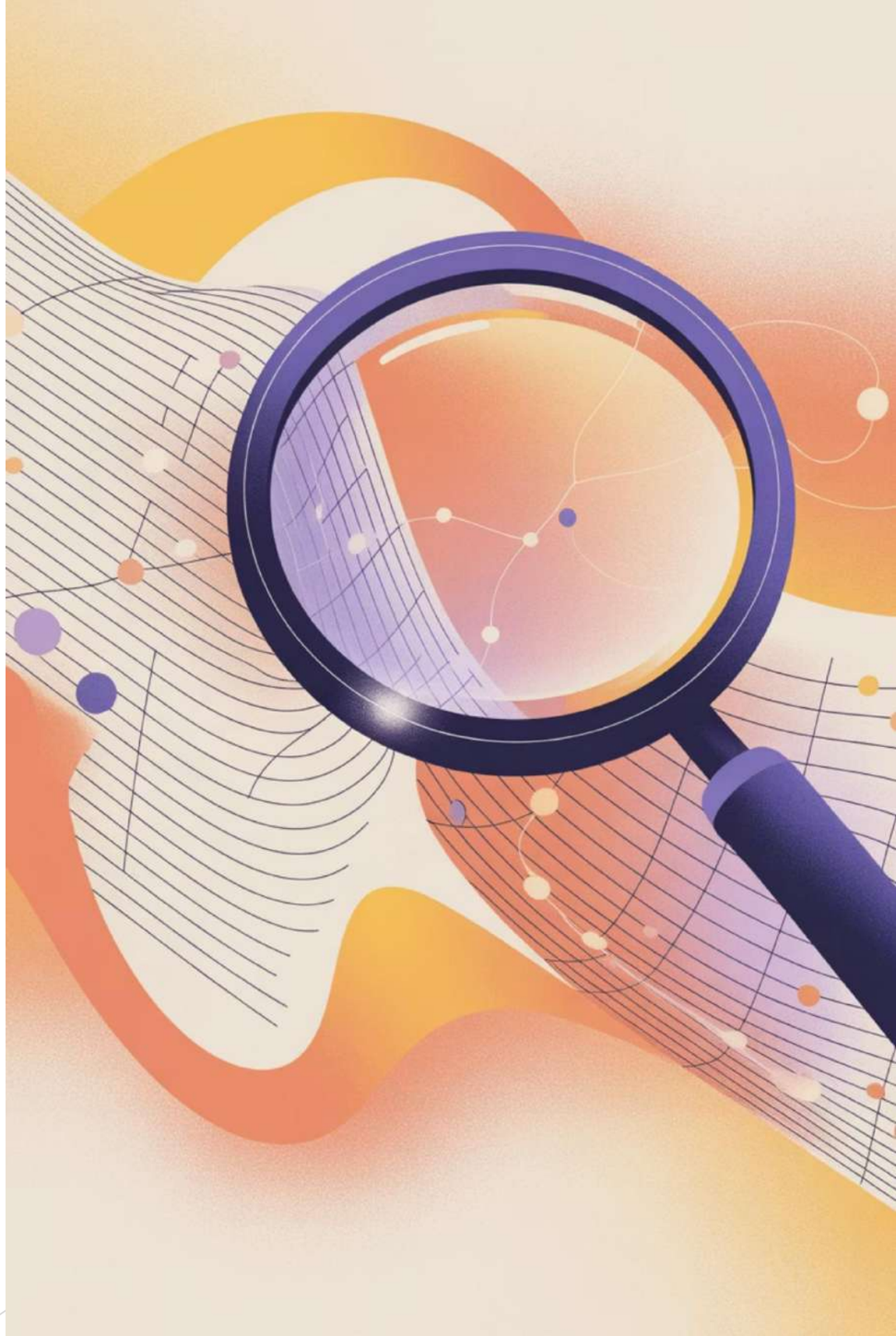
Fault Tolerance

Robust to partial data loss or errors, maintaining performance through distributed processing.

4

Proven Success

Demonstrated efficacy across diverse fields like fraud detection and medical diagnosis.



Real-World Applications of ANN Classification

Artificial Neural Networks provide critical classification capabilities across numerous industries, driving efficiency and informed decisions.

Fraud Detection

Identifies anomalous transaction patterns to flag potential fraud in financial services.

Medical Diagnosis

Assists in classifying medical images and patient data for accurate disease diagnosis.

Customer Behavior

Predicts customer churn, purchasing habits, and response to marketing campaigns.

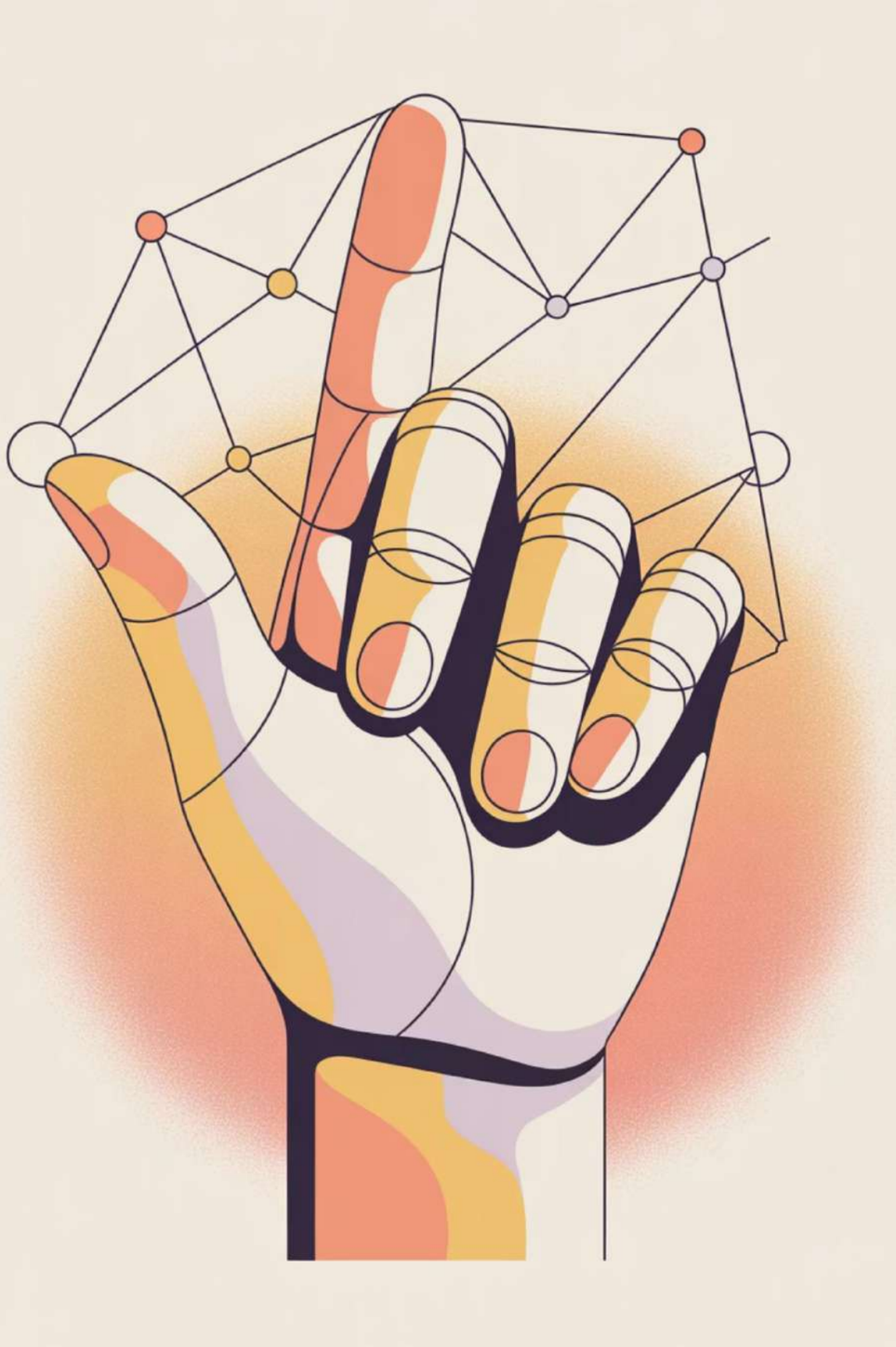
Bioinformatics

Classifies gene sequences and protein structures for advanced disease research.

Conclusion: ANN Empowers Smarter Data Mining

Artificial Neural Networks are indispensable for modern data mining, transforming raw data into strategic assets.

- ANNs significantly boost the accuracy of classification tasks within vast and intricate datasets.
- Their design, inspired by biological intelligence, offers unparalleled computational power for pattern recognition.
- Organizations leverage ANNs to extract actionable insights from data warehouses, driving informed decision-making.
- The evolution of data mining classification is increasingly shaped by sophisticated neural network models.



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